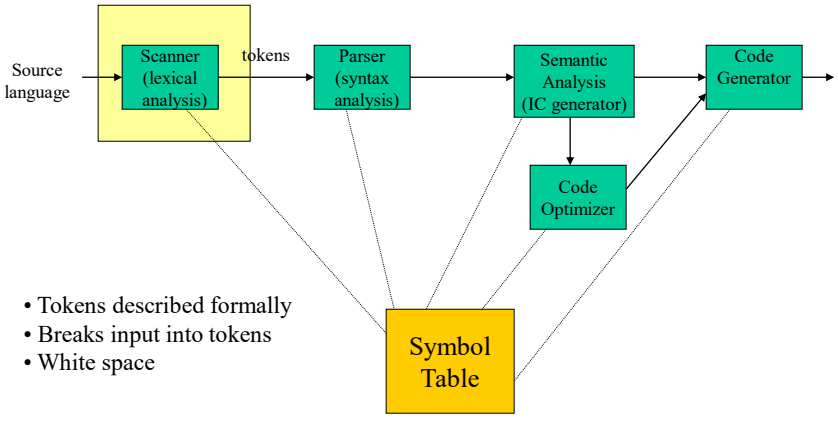
Lecture 2: Lexical Analysis

2nd 2019-2020

Lexical Analysis - Scanning

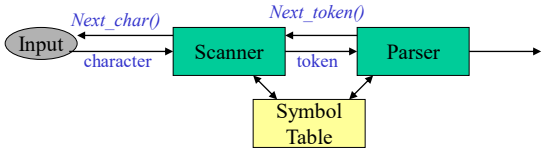


- Tokens described formally
- Breaks input into tokens
- White space

Lexical Analysis

INPUT: sequence of characters

OUTPUT: sequence of tokens



A lexical analyzer is generally a subroutine of parser:

- Simpler design
- Efficient
- Portable

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Definitions

- **token** – set of strings defining an atomic element with a defined meaning
- **pattern** – a rule describing a set of string
- **lexeme** – a sequence of characters that match some pattern

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- int x = 10
- Token:INT pattern: r = int lexeme:”int”
- Token:ID pattern: r = IDreg lexeme:”x”
- Token:ASSIGN pattern: r = ASSIGNreg lexeme:”=”
- Token:integer_num pattern: r = Integer_{reg} lexeme:”10”

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Examples

Token	**Pattern**	**Sample Lexeme**
while	while	while
relation_op	= | != | < | >	<
integer	(0-9)*	42
string	Characters between “ “	“hello”

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Input string: size := r * 32 + c

<token,lexeme> pairs:

- <id, size>
- <assign, :=>
- <id, r>
- <arith_symbol, *>
- <integer, 32>
- <arith_symbol, +>
- <id, c>

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Implementing a Lexical Analyzer

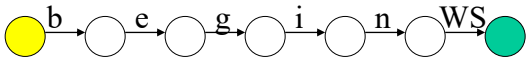
Practical Issues:

- Input buffering
- Translating RE into executable form
- Must be able to capture a large number of tokens with single machine
- Interface to parser
- Tools

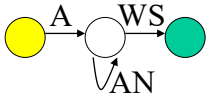
8

Capturing Multiple Tokens

Capturing keyword “begin”



Capturing variable names

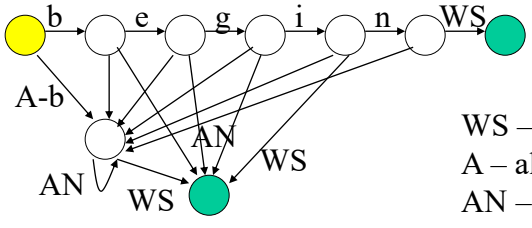


WS – white space
A – alphabetic
AN – alphanumeric

What if both need to happen at the same time?

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Capturing Multiple Tokens



WS – white space
A – alphabetic
AN – alphanumeric

Machine is much more complicated – just for these two tokens!

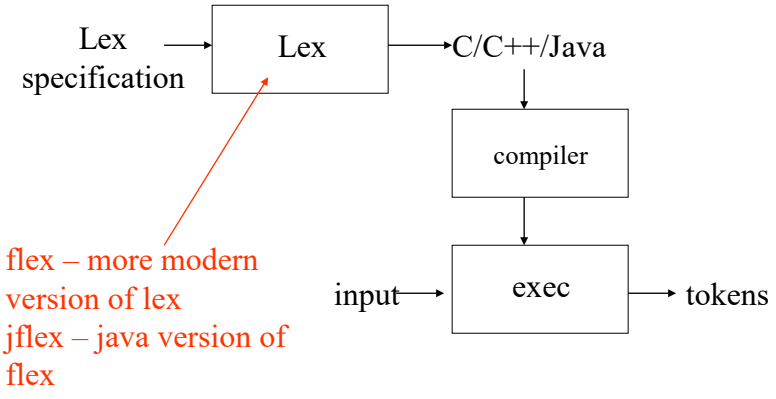
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Real lexer (handcoded)

- Comes from C# compiler in Rotor
- >950 lines of C++

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Lex – Lexical Analyzer Generator



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Lex Specification (flex)

```
%{
int charCount=0, wordCount=0, lineCount=0;
%}
word   [^ \t\n]+
%%
{word}      {wordCount++; charCount += strlen(yytext); }
[\n]        {charCount++; lineCount++;}
.           {charCount++;}
%%
main() {
   yylex();
   printf("Characters %d, Words: %d, Lines: %d\n",
charCount, wordCount, lineCount);
}
```

Definitions – Code, RE

Rules – RE/Action pairs

User Routines

Lex Specification (jflex)

```
import java.io.*;
%%
%class ex1
%unicode
%line
%column
%standalone
%{
static int charCount = 0, wordCount = 0, lineCount = 0;
public static void main(String [] args) throws IOException
{
    ex1 lexer = new ex1(new FileReader(args[0]));
    lexer.yylex();
    System.out.println("Characters: " + charCount +
            " Words: " + wordCount +" Lines: " +lineCount);
}
%}
%type Object    //this line changes the return type of yylex into Object
word = [^ \t\n]+
%%
{word}      {wordCount++; charCount += yytext().length(); }
[\n]        {charCount++; lineCount++; }
.           {charCount++; }
```

Definitions – Code, RE

Rules – RE/Action pairs

Lex definitions section

```
%{
int charCount=0, wordCount=0, lineCount=0;
%}
word   [^ \t\n]+
```

- C/C++/Java code:
  - Surrounded by %{… %} delimiters
  - Declare any variables used in actions
- RE definitions:
  - Define shorthand for patterns:

    digit [0-9]

    letter [a-z]

    *ident {letter}({letter}|{digit})**
  - Use shorthand in RE section: {*ident*}

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Lex Regular Expressions

```
{word}     {wordCount++; charCount += strlen(yytext); }
[\n]       {charCount++; lineCount++;}
.          {charCount++;}
```

- Match explicit character sequences
  - integer, “+++”, \<\>
- Character classes
  - [abcd]
  - [a-zA-Z]
  - [^0-9] – matches non-numeric

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- Alternation
  - twelve | 12
- Closure
  - * - zero or more
  - + - one or more
  - ? – zero or one
  - {*number*}, {*number*,*number*}

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- Other operators
  - . – matches any character except newline
  - ^ - matches beginning of line
  - $ - matches end of line
  - / - trailing context
  - () – grouping
  - {} – RE definitions

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Lex Operators

Highest: closure
concatenation
alternation



Special lex characters:

- \ / * + > “ { } . $ ( ) | % [ ] ^

Special lex characters inside [ ]:

- \ [ ] ^

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Examples

- a.*z
- (ab)+
- [0-9]{1,5}
- (ab|cd)?ef = *abef,cdef,ef*
- -?[0-9]\.[0-9]

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Lex Actions

Lex actions are C (C++, Java) code to implement some required functionality

- Default action is to echo to output
- Can ignore input (empty action)
- ECHO – macro that prints out matched string
- *yytext* – matched string
- *yyleng* – length of matched string (not all versions have this)

In Java:
yytext() and
yytext().length()

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User Subroutines

```
main() {
   yylex();
   printf("Characters %d, Words: %d, Lines: %d\n",charCount,
wordCount, lineCount);
}
```

- C/C++/Java code
- Copied directly into the lexer code
- User can supply 'main' or use default

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How Lex works

Lex works by processing the file one character at a time, trying to match a string starting from that character.

1. Lex *always* attempts to match the longest possible string.
2. If two rules are matched (and match strings are same length), the first rule in the specification is used.

Once it matches a string, it starts from the character after the string.

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Lex Matching Rules

1. Lex *always* attempts to match the longest possible string.

```
beg     {…}
begin   {…}
in      {…}
```

Input 'begin' can match either of the first two rules.
The second rule will be chosen because of the length.

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Lex Matching Rules

2. If two rules are matched (the matched strings are same length), the first rule in the specification is used.

```
begin      {… }
[a-z]+     {…}
```

Input ‘begin’ can match both rules – the first one will be chosen

Lex Example: Extracting white space

```
%{
#include <stdio.h>
%}
%%
[ \t\n]              ;
.                    {ECHO;}
%%
```

To compile and run above (simple.l):

```
flex simple.l                    flex simple.l
gcc lex.yy.c –ll                 g++ -x c++ lex.yy.c –ll
a.out < input                    a.out < input
```

-lfl on some systems

Lex Example: Extracting white space (Java)

```
%%
%class ex0
%unicode
%line
%column
%standalone
%%
[^ \t\n]      {System.out.print(yytext());}
.                     {}
[\n]                  {}
```

name of class to build

To compile and run above (simple.l):

```
java -jar ~cs540/JFlex.jar simple.l
javac ex0.java
java ex0 inputfile
```

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Input:

This is a file

of stuff we want to extract all

white space from

Output:

Thisisafileofstuffwewantoextractallwhitespacefrom

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Lex (C/C++)

- Lex always creates a file ‘lex.yy.c’ with a function yylex()
- -ll directs the compiler to link to the lex library (-lfl on some systems)
- The lex library supplies external symbols referenced by the generated code
- The lex library supplies a default main:
  main(int ac,char **av) {return yylex(); }

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Lex Example 2: Unix wc

```
%{ int charCount=0, wordCount=0, lineCount=0;
%}
word   [^ \t\n]+
%%
{word}   {wordCount++; charCount += strlen(yytext); }
[\n]        {charCount++; lineCount++;}
.           {charCount++;}
%%
main() {
   yylex();
   printf("Characters %d, Words: %d, Lines: %d\n",charCount,
    wordCount, lineCount);
}
```

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Lex Example 3: Extracting tokens

```
%%
and                     return(AND);
array                   return(ARRAY);
begin                   return(BEGIN);
.
.
.
\[                      return('[');
":="                    return(ASSIGN);
[a-zA-Z][a-zA-Z0-9_]*   return(ID);
[+-]?[0-9]+             return(NUM);
[ \t\n]                 ;
%%
```

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Uses for Lex

- **Transforming Input** – convert input from one form to another (example 1). *yylex*() is called once; return is not used in specification
- **Extracting Information** – scan the text and return some information (example 2). *yylex*() is called once; return is not used in specification.
- **Extracting Tokens** – standard use with compiler (example 3). Uses return to give the next token to the caller.

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Lex States

- Regular expressions are compiled to state machines.
- Lex allows the user to explicitly declare multiple states.

  %s COMMENT
- Default initial state INITIAL (0)
- Actions for matched strings may be different for different states

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Lex States

```
%{
int ctr = 0;
int linect = 1;
%}
%s COMMENT
%%
<INITIAL>.              ECHO;
<INITIAL>[\n]           {linect++; ECHO;}
<INITIAL>"/*"           {BEGIN COMMENT; ctr = 1;}
<COMMENT>.              ;
<COMMENT>[\n]           linect++;
<COMMENT>"*/"           {if (ctr == 1) BEGIN INITIAL;
                         else ctr--;
                        }
<COMMENT>"/*"           {ctr++;}
%%
```

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Project #1: Using Lex

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